Notebook II, Gate 2 - Horizontal Screening Jet and Deformation Intertwiner

Horizontal Finite-ε Screening Jet and Deformation Intertwiner

Notebook II, Gate 2 - exact resonant source families, fixed-fibre PBW comparison of the moving targets, regulated opposite-polarization BCH conversion, periodic coefficient extraction, and the represented A_1 connection equation.

GATE 2 CLOSED OUTCOME B AT A1 NOT AN ORDINARY SECOND SCREENING

Verdict. For each A_1 root component and each integral monodromy label, the two regular parent screenings form exact intertwiners on horizontal resonant Fock families. Canonical PBW-flat source and target frames identify their moving Hom-spaces with one fixed leading Hom-space. After the scalar, z-independent anti-normal-ordering subtraction, the odd numerator is genuinely divisible by $t = \varepsilon^2$, and its coefficient $Q^{[1]}$ is finite, nonzero and linearly independent from $Q^{[0]}$. The two target representations nevertheless follow different momentum curves. Their first represented-current coefficients do not agree, so $Q^{[1]}$ obeys an inhomogeneous connection/deformation-intertwiner equation and is not an ordinary screening of the frozen Carrollian algebra in the PBW-flat frame. Gate 2 therefore closes with Outcome B at A_1. Regulator and base-point changes are scalar triangular; general connection changes obey the full source-target gauge law. No quantum screening doublet and no nonlinear A_2/W_3 theorem is claimed.

The Gate-1 local result remains true: the bounded density symbol $(\beta_1^{\mathrm{sf}}\varphi^- + B_{\mathrm{diff}}\varphi^+)E$ has remainder $(\beta_1^{\mathrm{sf}})^2/4$. Gate 2 does not erase that calculation. It shows why the local criterion was insufficient: the exact plane-to-cylinder one-form carries an affine logarithmic factor, and the barred parent normal ordering becomes anti-normal ordering. The parent-coefficient residue contains both effects and is not the residue of the truncated local symbol.

1. Frozen input and notation

The immutable upstream artifact is `Notebook_II_Gate1_Freeze_Grade_Peer_Review_Corrected.zip`. Its SHA-256 is:

`860df721f9de6c266ff4f9016aedce6c1bc908afa66ec36095ec3094de42c871`

Gate 2 uses its constrained Cartan projector $\kappa_{AB} = \delta_{AB} - 1/N$, crossed brackets

$$[J^+_{A,m}, J^-_{B,n}] = \frac{m}{2}\kappa_{AB}\delta_{m+n,0}, \qquad [p^-_A, q^+_B] = [p^+_A, q^-_B] = \frac{1}{2}\kappa_{AB},$$

and the dictionary $\beta_1^{\mathrm{sf}} = -2\mu_1$. The detailed operator proof below is written in the $N = 2$ single-field normalization, where $\varphi^{\pm}_{\mathrm{sf}} = -\alpha_1 \cdot \varphi^{\pm}/2$, $(\alpha_1, \alpha_1) = 2$, and therefore

$$[J^+_m, J^-_n] = \frac{m}{4}\delta_{m+n,0}, \qquad [p^-_{\mathrm{sf}}, q^+_{\mathrm{sf}}] = [p^+_{\mathrm{sf}}, q^-_{\mathrm{sf}}] = \frac{1}{4}.$$

The root form is recovered by replacing $(\beta_1^{\mathrm{sf}})^2/2$ with $\mu_1^2(\alpha_i, \alpha_i)$.

Definition 1.1 - exact regular branches

Put $t = \varepsilon^2$, $\alpha(t) = \alpha^+ + t\alpha^-$, and $\bar\alpha(t) = \alpha^+ - t\alpha^-$. Let $b(t), \bar b(t) \in \mathbb{C}[[t]]$ be the unique regular solutions with the same nonzero constant term $b_1 = -2/\alpha^+$ of

$$tb(t)^2 - 2\alpha(t)b(t) - 4 = 0, \qquad t\bar b(t)^2 - 2\bar\alpha(t)\bar b(t) - 4 = 0.$$

Writing $b = b_1 + tb_3 + \cdots$, $\bar b = b_1 + t\bar b_3 + \cdots$ gives

$$\alpha^+ b_3 = \frac{b_1^2}{2} - \alpha^- b_1, \qquad \alpha^+ \bar b_3 = \frac{b_1^2}{2} + \alpha^- b_1.$$

2. Common completed source and target category

Definition 2.1 - horizontal Fock families and fixed formal neighborhood

Let $R = \mathbb{C}[[t]]$. For a formal momentum curve $\lambda(t) = (\lambda^+(t), \lambda^-(t))$, let

$$\mathcal{F}_R(\lambda(t)) = \mathcal{P}_{\mathrm{fin}} \otimes R|\lambda(t)\rangle, \qquad \widehat{\mathcal{F}}_R(\lambda(t)) = \widehat{\mathcal{P}}_E \otimes R|\lambda(t)\rangle, \quad \widehat{\mathcal{P}}_E = \prod_{E \geq 0} \mathcal{P}[E].$$

The zero modes $p^\pm$ act diagonally on the vacuum and the negative oscillators give a PBW basis. The canonical PBW-flat trivialization

$$\tau_\lambda : X|\lambda(t)\rangle \longmapsto X|\lambda(0)\rangle, \qquad X \in \mathcal{P}_{\mathrm{fin}},$$

identifies each moving Fock family coefficientwise with its leading fibre. Requiring preservation of every negative-mode PBW monomial and of a chosen normalized vacuum *vector* fixes this trivialization uniquely. If only the vacuum line is retained and its normalization is relaxed, scalar frame freedom remains.

For displaying derivatives of sector translations, but *not* as the screening source, use the fixed-lattice formal neighborhood

$$\mathscr{N}_{\Lambda_0} = \mathbb{C}[\Lambda_0]\widehat{\otimes}\mathbb{C}[[t, \eta^+, \eta^-]], \qquad p^\pm = \lambda_0^\pm + \eta^\pm, \qquad q^\mp = -\frac{1}{2}\partial^{\mathfrak{h}}_{\eta^\pm}.$$

Thus $e^{\gamma q^+} f(\eta^-) = f(\eta^- - \kappa\gamma/2)$, and every moving translation factors exactly as

$$e^{\gamma(t)q^+} = e^{\gamma(0)q^+} e^{(\gamma(t)-\gamma(0))q^+}, \qquad e^{(\gamma(t)-\gamma(0))q^+} = 1 + O(t).$$

The first factor shifts the discrete leading lattice; the second is a t-adically small translation generated by the now explicit unexponentiated q^+. This replaces the invalid moving direct-sum group algebra: distinct moving group-algebra basis vectors are never declared equal modulo t.

Proposition 2.2 - common fixed-fibre Hom-space

Let $\lambda_m(t)$ be the horizontal source curve of Proposition 3.2. Since $[p^-_{\mathrm{sf}}, q^+_{\mathrm{sf}}] = [p^+_{\mathrm{sf}}, q^-_{\mathrm{sf}}] = 1/4$, the zero-mode exponential $e^{b(q^+ + tq^-)}$ shifts (p^+, p^-) by $(tb/4, b/4)$. Hence in the single-field normalization the two vertex shifts are

$$\Delta_{\mathrm{u}}(t) = \frac{1}{4}\big(tb(t), b(t)\big), \qquad \Delta_{\mathrm{b}}(t) = \frac{1}{4}\big(-t\bar{b}(t), \bar{b}(t)\big), \qquad \Delta_{\mathrm{u}}(0) = \Delta_{\mathrm{b}}(0) =: \Delta_0.$$

Rootwise, the minus component is $\mu(t)\kappa\alpha_i/2$ and the frozen dictionary is $\mu = -b/2$. At A_1, reducing with $p_{\mathrm{sf}} = -\alpha_1 \cdot p/2$ and $(\alpha_1, \alpha_1) = 2$ gives, consistently,

$$\Delta p^-_{\mathrm{sf}} = -\frac{1}{2}\alpha_1 \cdot \Big(\frac{\mu}{2}\alpha_1\Big) = -\frac{\mu}{2} = \frac{b}{4}.$$

The two target trivializations

$$\tau_{\mathrm{u},m} : \widehat{\mathcal{F}}_R(\lambda_m + \Delta_{\mathrm{u}}) \to \widehat{\mathcal{F}}(\lambda_m(0) + \Delta_0)[[t]],$$
$$\tau_{\mathrm{b},m} : \widehat{\mathcal{F}}_R(\lambda_m + \Delta_{\mathrm{b}}) \to \widehat{\mathcal{F}}(\lambda_m(0) + \Delta_0)[[t]]$$

and the source trivialization $\tau_{\mathrm{s},m}$ place both parent intertwiners in the same complete module

$$\mathscr{H}_m = \mathrm{Hom}_{\mathbb{C}}\big(\mathcal{F}(\lambda_m(0)), \widehat{\mathcal{F}}(\lambda_m(0) + \Delta_0)\big)[[t]].$$

Creation exponentials are coefficientwise finite in energy, annihilation exponentials truncate on $\mathcal{P}_{\mathrm{fin}}$, and the PBW maps are continuous. Hence the renormalized ordered vertices define continuous elements of $\mathscr{H}_m\{z\}[\ell]$. The source of the screening theorem is the direct sum of the exact horizontal fibres over $m \in \mathbb{Z}$; the ambient neighborhood $\mathscr{N}_{\Lambda_0}$ is used only to write connection derivatives and to audit off-horizontal monodromy.

3. Complete oscillator and zero-mode embeddings

Proposition 3.1 - full Weyl embeddings

$$I_t(a_{A,n}) = t^{-1/2}J^+_{A,n} + t^{1/2}J^-_{A,n}, \quad I_t(\bar a_{A,n}) = t^{-1/2}J^+_{A,-n} - t^{1/2}J^-_{A,-n},$$

$$I_t(p) = t^{-1/2}p^+ + t^{1/2}p^-, \quad I_t(q) = t^{-1/2}q^+ + t^{1/2}q^-,$$
$$I_t(\bar p) = t^{-1/2}p^+ - t^{1/2}p^-, \quad I_t(\bar q) = -t^{-1/2}q^+ + t^{1/2}q^-.$$

They preserve both parent Weyl algebras and make their images commute. Because the formulas contain $t^{\pm 1/2}$, these embeddings are formed after scalar extension from $R = \mathbb{C}[[t]]$ to $K = \mathbb{C}((t^{1/2}))$. The transported screening vertices and contracted currents used below lie in the regular t-adic subcategory. The plane zero modes are obtained by restoring the Feigin-Fuchs shifts $a_0 = \mathcal{J}_0 + \alpha_{\rm pl}/2$ and $\bar a_0 = \bar{\mathcal{J}}_0 + \bar\alpha_{\rm pl}/2$; these shifts are not absorbed into $p^\pm$.

Proposition 3.2 - simultaneous monodromy section

For an integer Fourier label m, define $\lambda^\pm_m(t)$ by

$$\lambda^+_m = \frac{1}{2}\left(\frac{m-\delta}{b} + \frac{m-\bar\delta}{\bar b}\right), \qquad t\lambda^-_m = \frac{1}{2}\left(\frac{m-\delta}{b} - \frac{m-\bar\delta}{\bar b}\right),$$

where $\delta = tb^2/4$, $\bar\delta = t\bar b^2/4$. Since $b - \bar b = O(t)$, both λ^+_m and λ^-_m are regular series. They solve simultaneously

$$\delta + b\lambda^+_m + tb\lambda^-_m = m, \qquad \bar\delta + \bar b\lambda^+_m - t\bar b\lambda^-_m = m.$$

Thus both parent one-forms have the same integral monodromy on one exact horizontal section of the momentum bundle.

Corollary 3.3 - no off-horizontal screening jets

If one perturbs the source by $p^+ = \lambda^+_m + \xi$, $p^- = \lambda^-_m + \nu$, the residual exponents are

$$a_{\rm u} = b\xi + tb\nu, \qquad a_{\rm b} = \bar b\xi - t\bar b\nu.$$

For generic $t, b, \bar b$, simultaneous resonance forces $\xi = \nu = 0$. Therefore arbitrary momentum jets are not objects of the finite-parent screening domain. This restriction is essential: the formal neighborhood of Definition 2.1 compares moving target sectors, whereas the source is the horizontal fibre itself.

Cell 1 - branch equations and common monodromy

```
import sympy as sp
t, ap, am, b1, b3, bb3, m = sp.symbols('t alpha_plus alpha_minus b1 b3 bbar3 m', nonzero=True)
b = b1+t*b3; bb = b1+t*bb3
F = sp.expand(t*b**2-2*(ap+t*am)*b-4)
Fb = sp.expand(t*bb**2-2*(ap-t*am)*bb-4)
assert sp.expand(F).coeff(t,0) == -2*ap*b1-4
assert sp.expand(F).coeff(t,1) == b1**2-2*ap*b3-2*am*b1
assert sp.expand(Fb).coeff(t,1) == b1**2-2*ap*bb3+2*am*b1
d, db = t*b**2/4, t*bb**2/4
lp = sp.Rational(1,2)*((m-d)/b+(m-db)/bb)
tlm = sp.Rational(1,2)*((m-d)/b-(m-db)/bb)
assert sp.simplify(d+b*lp+b*tlm-m) == 0
assert sp.simplify(db+bb*lp-bb*tlm-m) == 0
xi, nu = sp.symbols('xi nu')
au, ab = b*xi+t*b*nu, bb*xi-t*bb*nu
assert sp.simplify(sp.det(sp.Matrix([[b,t*b],[bb,-t*bb]]))+2*t*b*bb) == 0
assert sp.solve([au,ab],[xi,nu], dict=True) == [{xi:0,nu:0}]
print('regular horizontal branches pass; off-horizontal simultaneous jets are excluded')
```

4. Exact transported finite-t vertices

Lemma 4.1 - affine one-form factor

The plane-cylinder potential relations of Gate 1 and the source-plane signs $\beta_{\rm pl} = -\sqrt{t}\,b(t)$, $\bar\beta_{\rm pl} = +\sqrt{t}\,\bar b(t)$ give

$$\mathbb{V}_t(z) = z^{\delta(t)} :\exp[b(t)(\varphi^+(z) + t\varphi^-(z))]:,$$
$$\bar{\mathbb{V}}^{\rm raw}_{t,\rho}(z) = z^{\bar\delta(t)} \underset{\rho}{\infty} \exp[\bar b(t)(\varphi^+(z) - t\varphi^-(z))] \underset{\rho}{\infty},$$

with $\delta = 1 + \alpha(t)b/2$ and $\bar\delta = 1 + \bar\alpha(t)\bar b/2$. The weight-one equations imply the exact identities

$$\boxed{\delta(t) = \frac{tb(t)^2}{4}, \qquad \bar{\delta}(t) = \frac{t\bar{b}(t)^2}{4}.}$$

The z^δ factors are part of the transported one-forms. Omitting them reproduces the Gate-1 local obstruction and is not an exact finite-parent comparison.

Corollary 4.2 - first coefficients before barred subtraction

With $E =: e^{b_1\varphi^+} :$,

$$\mathbb{V}_t = E + t : \left(b_1\varphi^- + b_3\varphi^+ + \frac{b_1^2}{4}\ell \right) E : + O(t^2),$$

$$\bar{\mathbb{V}}_t^{\mathrm{raw}} = E + t \infty \left(-b_1\varphi^- + \bar{b}_3\varphi^+ + \frac{b_1^2}{4}\ell \right) E \infty + O(t^2).$$

5. Opposite-polarization BCH conversion

Theorem 5.1 - regulated conversion factor

Insert a radial damping $0 < \rho < 1$ in each barred contraction. For one A_1 field the annihilation and creation exponents A_ρ, B_ρ obey

$$[A_\rho, B_\rho] = \frac{t\bar{b}(t)^2}{2} \sum_{n\geq 1} \frac{\rho^n}{n}, \mathbf{1} = -\frac{t\bar{b}(t)^2}{2} \log(1-\rho)\, \mathbf{1}.$$

Consequently

$$e^{A_\rho} e^{B_\rho} = \bar{Z}_{t,\rho}\, e^{B_\rho} e^{A_\rho}, \qquad \boxed{\bar{Z}_{t,\rho} = (1-\rho)^{-t\bar{b}(t)^2/2}}, \qquad Z_t = 1.$$

The factors are scalar, z-independent, and coefficientwise analytic in t at fixed ρ. Define the minimal renormalized barred field by $\widehat{\bar{V}}_t = \lim_{\rho\uparrow 1} \bar{Z}_{t,\rho}^{-1} \bar{\mathbb{V}}_{t,\rho}^{\mathrm{raw}}$. A finite subtraction multiplies it by $e^{c(t)}$, $c(t) \in t\mathbb{C}[[t]]$. For a root vertex the exponent $t\bar{b}^2/2$ is $t\bar{\mu}^2(\alpha_i, \alpha_i)$.

Cell 2 - BCH coefficient and scalar character

```
import sympy as sp
n, t, b, rho = sp.symbols('n t b rho', positive=True)
# [J+_n-tJ-_n, J+_-n-tJ-_-n] = -t*n/2 in the A1 single-field normalization.
bracket = -t*n/sp.Integer(2)
coeff = sp.simplify((-b/n)*(b/n)*bracket)
assert coeff == t*b**2/(2*n)
H = -sp.log(1-rho)
Z = sp.exp(t*b**2*H/2)
assert sp.diff(sp.log(Z), sp.Symbol('z')) == 0
print('barred BCH factor = exp(t*b^2*H_rho/2), scalar and z-independent')
```

6. Periodic logarithmic residue

Definition 6.1 - based logarithmic residue

Fix $z_0 \in \mathbb{C}^\times$ and the branch $z = z_0 e^{i\theta}$, $0 \leq \theta \leq 2\pi$, $\ell = \log(z/z_0) = i\theta$. Define continuously on logarithmic Laurent fields

$$\operatorname*{LRes}_{z_0} f := \frac{1}{2\pi i} \int_0^{2\pi} f(z_0 e^{i\theta})\, i z_0 e^{i\theta} d\theta.$$

Equivalently, for $r \in \mathbb{Z}$ and formal a,

$$\boxed{\operatorname*{LRes}_{z_0} \left(z^{r-1} e^{a\ell} \right) = z_0^r \frac{e^{2\pi i a} - 1}{2\pi i (r+a)},}$$

with the removable value at $r + a = 0$. It reduces to the usual residue at $a = 0$, but $\mathrm{LRes}_{z_0}(d\ell) = 1$. This is why a local total derivative cannot be discarded before the branch prescription.

Proposition 6.2 - horizontal periodicity and the boundary term

For a weight-one field with residual exponent a, the based integral of the total derivative is

$$\operatorname*{LRes}_{z_0} \partial_z\big(z^{n+1}V(z)\big) = \frac{e^{2\pi i a}-1}{2\pi i}\, z_0^{n+1}V(z_0).$$

This off-horizontal logarithmic boundary anomaly is generically nonzero on the ambient neighborhood. On the exact source curves of Proposition 3.2, however, $a_{\mathrm{u}}(t) = a_{\mathrm{b}}(t) = 0$ identically as formal series, not merely at $t = 0$. Both boundary terms therefore vanish before coefficient extraction. The two finite-parent cylinder integrals are ordinary periodic screening modes on their horizontal fibres, and coefficient extraction in t commutes with the integral in the energy-product topology.

The logarithmic formula is consequently a derived representative of the primary horizontal parent-family prescription. Its separate logarithmic terms must be summed before applying the boundary test; differentiating the exact resonance equation cancels their first-order monodromy anomaly.

Cell 3 - logarithmic residue expansion

```
import sympy as sp
a, z0 = sp.symbols('a z0'); r = sp.symbols('r', integer=True, nonzero=True)
Rr = z0**r*(sp.exp(2*sp.pi*sp.I*a)-1)/(2*sp.pi*sp.I*(r+a))
assert sp.simplify(sp.limit(Rr,a,0)) == 0
assert sp.simplify(sp.diff(Rr,a).subs(a,0)-z0**r/r) == 0
R0 = (sp.exp(2*sp.pi*sp.I*a)-1)/(2*sp.pi*sp.I*a)
assert sp.simplify(sp.limit(R0,a,0)) == 1
assert sp.simplify(sp.limit(sp.diff(R0,a),a,0)-sp.pi*sp.I) == 0
boundary = (sp.exp(2*sp.pi*sp.I*a)-1)/(2*sp.pi*sp.I)
assert boundary.subs(a,0) == 0
assert sp.diff(boundary,a).subs(a,0) == 1
print('based residue verified: off-horizontal anomaly nonzero, horizontal boundary exactly zero')
```

7. Coefficient operators

Definition 7.1 - renormalized parent charges and channels

$$\widehat{Q}_t = 2\pi i\,\mathrm{LRes}_{z_0}\mathbb{V}_t, \qquad \widehat{\bar{Q}}_t = 2\pi i\,\mathrm{LRes}_{z_0}\widehat{\bar{V}}_t,$$

These first act between their moving horizontal Fock families. Put them in the common fixed fibre by

$$\mathscr{Q}_{\mathrm{u}}(t) = \tau_{\mathrm{u},m}\widehat{Q}_t\tau_{\mathrm{s},m}^{-1}, \qquad \mathscr{Q}_{\mathrm{b}}(t) = \tau_{\mathrm{b},m}\widehat{\bar{Q}}_t\tau_{\mathrm{s},m}^{-1}.$$
$$Q^{(0)}(t) = \frac{\mathscr{Q}_{\mathrm{u}}(t)+\mathscr{Q}_{\mathrm{b}}(t)}{2}, \qquad Q^{(1)}(t) = \frac{\mathscr{Q}_{\mathrm{u}}(t)-\mathscr{Q}_{\mathrm{b}}(t)}{2t}.$$

Set $Q^{[0]} = \mathrm{Coeff}_{t^0}Q^{(0)}(t)$ and $Q^{[1]} = \mathrm{Coeff}_{t^0}Q^{(1)}(t)$. The superscript in square brackets is a coefficient/flag label, not a mode number.

Theorem 7.2 - finiteness, source/target, and nonvanishing

Both trivialized charges belong to the same topologically free module $\mathscr{H}_m$. Their constant terms are the same leading vertex map. Equality of the constant coefficients in this fixed PBW basis means, now legitimately, that $\mathscr{Q}_{\mathrm{u}} - \mathscr{Q}_{\mathrm{b}} \in t\mathscr{H}_m$. Therefore $Q^{[1]}$ is finite and continuous, with

$$Q^{[1]} : \mathcal{F}(\lambda_m(0)) \longrightarrow \widehat{\mathcal{F}}\left(\lambda_m(0) + \frac{\mu_1}{2}\kappa\alpha_i\right), \qquad \mu_1 = -\frac{1}{2}b_1.$$

In the canonical affine formal-neighborhood gauge, its density representative begins

$$\boxed{\mathcal{S}^{[1]}_{\mathrm{odd}} =: \big(b_1\varphi^- + B_{\mathrm{diff}}\varphi^+\big)E :,}$$

where $B_{\mathrm{diff}} = (b_3 - \bar{b}_3)/2 = -\alpha^- b_1/\alpha^+$. Both q^+ and q^- in this representative are the explicit connection derivatives of Definition 2.1; they are not undeclared operators on a moving group algebra.

Nonvanishing does not rely on an off-resonant source jet. Take the horizontal vacuum with $m = -2$. In the coefficient of z^{-1}, the projection of $Q^{[1]}|\lambda_{-2}(0)\rangle$ onto the PBW vector $J^-_{-1}|\lambda_{-2}(0)+\Delta_0\rangle$ is proportional to b_1, hence is nonzero. The corresponding projection of $Q^{[0]}$ contains only the J^+_{-1} creation direction. Thus $Q^{[1]} \neq 0$ and the two coefficient maps are linearly independent.

Cell 4 - fixed-target divisibility and PBW nonvanishing witness

```
import sympy as sp
ap, am, b1 = sp.symbols('alpha_plus alpha_minus b1', nonzero=True)
b3 = b1**2/(2*ap)-am*b1/ap
bb3 = b1**2/(2*ap)+am*b1/ap
B = sp.simplify((b3-bb3)/2); D = sp.simplify((b3+bb3)/2)
assert B == -am*b1/ap
assert D == b1**2/(2*ap)
t, eta = sp.symbols('t eta')
su, sb = b1/4+t*b3/4, b1/4+t*bb3/4
f = 1+2*eta+3*eta**2
Tu = sp.series(f.subs(eta,eta-(su-b1/4)),t,0,2).removeO()
Tb = sp.series(f.subs(eta,eta-(sb-b1/4)),t,0,2).removeO()
assert sp.expand(Tu-Tb).coeff(t,0) == 0
assert sp.simplify(sp.expand(Tu-Tb).coeff(t,1)+(b3-bb3)*sp.diff(f,eta)/4) == 0
Jm1 = sp.symbols('Jminus_1')
odd_level_one = b1*Jm1+B*sp.symbols('Jplus_1')
assert sp.expand(odd_level_one).coeff(Jm1) == b1
print('fixed-target difference is t-divisible; m=-2 PBW witness has nonzero Jminus_-1 coefficient')
```

8. The represented A_1 coefficient equation

Proposition 8.1 - universal currents versus represented actions

Let $T_{\mathrm{p}}(t)$, $\bar{T}_{\mathrm{p}}(t)$ be the two parent Feigin-Fuchs tensors with symmetric parent normal ordering, transported with the embeddings above. Define

$$M(t) = t\big(T_{\mathrm{p}}(t) + \bar{T}_{\mathrm{p}}(t)\big), \qquad L(t) = T_{\mathrm{p}}(t) - \bar{T}_{\mathrm{p}}(t).$$

Direct expansion of the universal fields, with the barred contribution converted before the limit, gives, up to commuting vacuum constants,

$$M(t) = M^{[0]} + t^2 M^{[2]}, \qquad L(t) = L^{[0]},$$
$$M^{[0]} = 2J^+J^+ + 2\alpha^+\partial J^+, \quad L^{[0]} = 4\langle J^+J^-\rangle_{\mathrm{av}} + 2\alpha^+\partial J^- + 2\alpha^-\partial J^+.$$

Thus the universal noncentral current formulas have no explicit order-t coefficient. This statement does *not* set the first coefficient of the PBW-trivialized source and target representations to zero: their zero-mode eigenvalues move with their momentum curves.

Theorem 8.2 - PBW-flat deformation-intertwiner equations

On each horizontal source fibre, the own-factor weight-one commutator is the boundary term of Proposition 6.2 and vanishes because $a_{\mathrm{u}} = a_{\mathrm{b}} = 0$ identically. The cross-factor commutators vanish because the two embedded Weyl algebras commute. The scalar BCH subtraction does not change an intertwining equation. Hence, before taking any coefficient,

$$[\widehat{Q}_t, L(t)] = [\widehat{Q}_t, M(t)] = [\widehat{\bar{Q}}_t, L(t)] = [\widehat{\bar{Q}}_t, M(t)] = 0.$$

Write the PBW-trivialized source and target actions separately as $U_{\mathrm{S}}(t)$, $U_{\mathrm{T,u}}(t)$, and $U_{\mathrm{T,b}}(t)$. Here $[Q, U] = U_{\mathrm{T}}Q - QU_{\mathrm{S}}$ is intertwiner notation. Applying the PBW maps preserves the exact equations but does not freeze the represented zero-mode eigenvalues. The two target paths are

$$p^+_{\mathrm{T,u}} = \lambda^+_m + \frac{tb}{4}, \qquad p^+_{\mathrm{T,b}} = \lambda^+_m - \frac{t\bar{b}}{4},$$
$$p^-_{\mathrm{T,u}} = \lambda^-_m + \frac{b}{4}, \qquad p^-_{\mathrm{T,b}} = \lambda^-_m + \frac{\bar{b}}{4}.$$

Their relative first tangent is therefore

$$\boxed{\Delta p^+ = \frac{b_1}{2}, \qquad \Delta p^- = \frac{B_{\mathrm{diff}}}{2} = \frac{b_3 - \bar{b}_3}{4}.}$$

Using the mode formulas of [FH25, eqs. (2.15a)-(2.15b)], conjugating the leading currents through the PBW frames gives the target-action differences

$$\boxed{\Delta M^{[1]}_{\mathrm{T},n} = \frac{b_1}{2}\big(4J^+_n - 2\alpha^+\delta_{n,0}\big),}$$

$$\boxed{\Delta L^{[1]}_{\mathrm{T},n} = \frac{b_1}{2}\big(4J^-_n - 2\alpha^-\delta_{n,0}\big) + \frac{B_{\mathrm{diff}}}{2}\big(4J^+_n - 2\alpha^+\delta_{n,0}\big).}$$

The common source first correction cancels when the two exact order-t intertwining equations are subtracted. Dividing by two yields, for every $n \in \mathbb{Z}$,

$$\boxed{[Q^{[1]}, U^{[0]}_n]_{\mathrm{PBW}} + \frac{1}{2}\Delta U^{[1]}_{\mathrm{T},n} Q^{[0]} = 0, \qquad U = L, M.}$$

This is the correct connection/deformation-intertwiner equation. It is inhomogeneous even though the universal fields have no explicit order-t term. For M_0, with $p^+_0 = m/b_1$ and $\alpha^+ b_1 = -2$,

$$\Delta M^{[1]}_{\mathrm{T},0} = 2(m+1), \qquad \boxed{[Q^{[1]}, M_0]_{\mathrm{PBW}} = -(m+1)Q^{[0]}.}$$

On the $m = -2$ component used in Theorem 7.2, this becomes $[Q^{[1]}, M_0]_{\mathrm{PBW}} = Q^{[0]} \neq 0$. Likewise $\lambda^-_m(0) = \alpha^- m/(\alpha^+ b_1)$, while the leading target minus momentum is $p^-_{\mathrm{T},0} = \lambda^-_m(0) + b_1/4$. This gives $\Delta L^{[1]}_{\mathrm{T},0} = b_1^2/2$, hence $[Q^{[1]}, L_0]_{\mathrm{PBW}} = -b_1^2 Q^{[0]}/4 \neq 0$. Therefore the PBW-flat coefficient is nonzero but is not an ordinary screening of the limiting algebra. This proves Outcome B rather than Outcome A.

Cell 5 - represented A1 target connection and coefficient equation

```
import sympy as sp
t,ap,am,b1,m = sp.symbols('t alpha_plus alpha_minus b1 m', nonzero=True)
b3=b1**2/(2*ap)-am*b1/ap; bb3=b1**2/(2*ap)+am*b1/ap
B=sp.simplify((b3-bb3)/2)
# Frozen A1 Weyl normalization shifts (p+,p-) by (t*b/4,b/4).
dp,dm=b1/2,B/2
p0=m/b1; q0=am*m/(ap*b1)+b1/4
p,q=sp.symbols('p q')
hM=2*p**2-2*ap*p
hL=4*p*q-2*ap*q-2*am*p
dM=sp.simplify(sp.diff(hM,p).subs(p,p0)*dp)
dL=sp.simplify(sp.diff(hL,p).subs({p:p0,q:q0})*dp
               +sp.diff(hL,q).subs({p:p0,q:q0})*dm)
assert sp.simplify(dM.subs(b1,-2/ap)-2*(m+1)) == 0
assert sp.simplify(dL-b1**2/2) == 0
assert sp.simplify((-dM/2).subs({b1:-2/ap,m:-2})-1) == 0
assert sp.simplify(-dL/2+b1**2/4) == 0
# Abstract subtraction of the two represented order-t intertwining equations.
UT0,US0,Qu1,Qb1,Q0,dUT=sp.symbols('UT0 US0 Qu1 Qb1 Q0 dUT',commutative=False)
Q1=(Qu1-Qb1)/2
odd=sp.expand(UT0*Q1-Q1*US0+dUT*Q0/2)
expected=UT0*Q1-Q1*US0+dUT*Q0/2
assert sp.expand(odd-expected) == 0
print('Outcome B verified: target connection survives and the PBW-flat equation is inhomogeneous')
```

9. Regularization, base point, and triangular flag

Proposition 9.1 - regulator, base point, and connection covariance

A finite barred subtraction $e^{c_1 t + O(t^2)}$ changes the odd coefficient by $-c_1 Q^{[0]}/2$. Changing the logarithmic branch or the base point on a fixed horizontal monodromy component changes the coefficient by a componentwise scalar multiple of $Q^{[0]}$.

The PBW-flat connection is part of Definition 2.1. More generally, let the two branch frames change by $g_{\mathrm{s},r} = 1 + tA_{\mathrm{s},r} + O(t^2)$ and $g_{\mathrm{t},r} = 1 + tA_{\mathrm{t},r} + O(t^2)$, $r = \mathrm{u}, \mathrm{b}$, and define their relative half-differences $A_{\mathrm{s}} = (A_{\mathrm{s,u}} - A_{\mathrm{s,b}})/2$, $A_{\mathrm{t}} = (A_{\mathrm{t,u}} - A_{\mathrm{t,b}})/2$. The odd coefficient obeys the exact gauge law

$$Q^{[1]} \longmapsto Q^{[1]} + A_{\mathrm{t}} Q^{[0]} - Q^{[0]} A_{\mathrm{s}}.$$

A common frame change on both branches has zero relative half-difference and leaves $Q^{[1]}$ unchanged. The chosen normalized vacuum vectors make the canonical PBW-flat frames unique. If branch-relative vacuum normalization is relaxed while every PBW monomial and only the vacuum lines are preserved, $A_{\mathrm{s}}, A_{\mathrm{t}}$ are scalar; together with finite subtraction and base-point changes this gives

$$\boxed{Q^{[1]} \longmapsto Q^{[1]} + cQ^{[0]}}$$

A momentum-differential re-trivialization produces additional q^+- or q^--type terms according to the displayed general gauge law. Simultaneously, $\Delta U^{[1]}_{\mathrm{T}} \mapsto \Delta U^{[1]}_{\mathrm{T}} + 2[A_{\mathrm{t}}, U^{[0]}_{\mathrm{T}}]$ and $\Delta U^{[1]}_{\mathrm{S}} \mapsto \Delta U^{[1]}_{\mathrm{S}} + 2[A_{\mathrm{s}}, U^{[0]}_{\mathrm{S}}]$. Thus the general

equation $[Q^{[1]}, U^{[0]}] + \Delta U_{\mathrm{T}}^{[1]} Q^{[0]}/2 - Q^{[0]} \Delta U_{\mathrm{S}}^{[1]}/2 = 0$ is gauge covariant; the canonical construction has $\Delta U_{\mathrm{S}}^{[1]} = 0$. Such a re-trivialization changes the connection category and is not silently identified with a scalar scheme change or used to claim an ordinary screening on the original PBW-flat fibres. In the canonical affine formal-neighborhood gauge, B_{diff} is fixed. This separates: (i) the parameter correction $D\partial_{b_1} Q^{[0]}$; (ii) the BCH finite subtraction; (iii) scalar triangular flag mixing; and (iv) a genuine connection change.

10. Classification of the three first-order directions

Direction	Actual operator status after Gate 2	Coefficient
Strict odd channel	Nonzero horizontal connection/deformation coefficient $Q^{[1]}$; not an ordinary A_1 screening in the PBW-flat frame; regulator/base-point ambiguity is modulo $Q^{[0]}$, while general connection changes obey Proposition 9.1	$B_{\mathrm{diff}} = -\alpha^- b_1/\alpha^+$ in the canonical affine gauge
Local corrected / unbarred first coefficient	Horizontal coefficient intertwiner $Q_{\mathrm{u}}^{[1]} = Q^{[1]} + Q_{\mathrm{even}}^{[1]}$; its truncated local representative has $B_{\mathrm{loc}} = b_3$	$B_{\mathrm{loc}} = B_{\mathrm{diff}} + b_1^2/(2\alpha^+) = b_3$
Even parameter derivative	Not a third flag generator; it is the horizontal parameter tangent $D\partial_{b_1} Q^{[0]}$, the affine logarithmic contribution whose monodromy cancels on the exact section, and a possible triangular scalar	$D = (b_3 + \bar{b}_3)/2 = b_1^2/(2\alpha^+)$

The Gate-1 quotient by parameter derivatives is therefore realized by horizontal coefficient maps: the strict odd and local-corrected representatives determine the same rank-two coefficient flag after the even parameter tangent is removed. The flag is $\langle Q^{[0]} \rangle \subset \langle Q^{[0]}, Q^{[1]} \rangle$, but only $Q^{[0]}$ is an ordinary limiting screening. The second generator is a deformation intertwiner carrying the represented target-connection term of Theorem 8.2.

11. Gate-2 acceptance ledger

Criterion	Status	Witness
G2.1 common completed category	PASS	Fixed-lattice neighborhood and PBW-flat fixed-fibre Hom-space, Definitions 2.1-2.2
G2.2 oscillator and zero-mode Weyl embeddings	PASS	Propositions 3.1-3.2 and explicit exclusion of off-horizontal jets
G2.3 explicit BCH factors	PASS	Theorem 5.1; $Z_t = 1$, $\bar{Z}_{t,\rho} = (1-\rho)^{-t\bar{b}^2/2}$
G2.4 finite nonzero $Q^{[1]}$	PASS	Fixed-basis t-divisibility and the $m = -2$, J_{-1}^- PBW witness in Theorem 7.2
G2.5 dependence and connection covariance	PASS	Scalar regulator/base-point ambiguity and the separate general source-target connection gauge law, Proposition 9.1
G2.6 periodic logarithmic prescription	PASS	Definition 6.1 and exact horizontal cancellation of the boundary anomaly, Proposition 6.2
G2.7 mixed A_1 equations for L, M	PASS	Separate source/target action expansions and the inhomogeneous PBW-flat connection equations, including explicit M_0 and L_0 tests, Theorem 8.2
G2.8 three corrections classified	PASS	§§9-10

Authorized sequencing. Gate 2 is closed with Outcome B. Notebook III may use $Q_i^{[0]}$ as ordinary leading screenings and may transport $Q_i^{[1]}$ only as connection/deformation coefficients. It must derive the nonlinear $U_3^{\pm}$ source and target action families and their mixed inhomogeneous equations. The phrase "quantum screening doublet" is not authorized by Notebook II.

References and provenance

[**FH25**] S. Fredenhagen and L. Hörl, *Free-field construction of Carrollian W_N-algebras*, JHEP 02 (2026) 218, arXiv:2509.14441v3. Used for the symmetric contraction, mode reversal, averaged ordering, Miura currents and the exact A_1 stress-tensor contraction.

[**DDF26**] E. D'Arcy, A. Delfante and S. Fredenhagen, *Holographic realization of higher-spin Carrollian free fields*, arXiv:2604.27068v1. Context for the classical screening charges and the bulk target.

[**TK86**] A. Tsuchiya and Y. Kanie, *Fock Space Representations of the Virasoro Algebra - Intertwining Operators* -, Publ. RIMS Kyoto Univ. 22 (1986) 259-327.

[**HLZ11**] Y.-Z. Huang, J. Lepowsky and L. Zhang, *Logarithmic tensor category theory for generalized modules for a conformal vertex algebra, I-II*, arXiv:1012.4193 and arXiv:1110.1929. Used only as background for Hom-valued fields with powers of a formal logarithm; the based residue in §6 is defined directly here.

[**AR19**] D. Adamović and G. Radobolja, *Self-dual and logarithmic representations of the twisted Heisenberg-Virasoro algebra at level zero*, Commun. Contemp. Math. 21 (2019) 1850008, arXiv:1703.00531.

All Gate-2 formulas beyond the cited parent conventions are derivations in this notebook. The package verifier checks algebraic identities and document consistency; it does not certify novelty, analytic Hilbert-space closure, or the future nonlinear A_2/W_3 theorem.